Functional Abstraction

But this corrected use of Leibniz notation is ugly. We had to introduce extraneous symbols (q and $\dot{q}$) in order to indicate the argument position specifying the partial derivative. Nothing would change here if we replaced q and $\dot{q}$ by a and b.[3] We can simplify the notation by admitting that the partial derivatives of the Lagrangian are themselves new functions, and by specifying the particular partial derivative by the position of the argument that is varied

$$\frac{d}{dt}((\partial_2 L)(t, w(t), \frac{d}{dt}w(t))) - (\partial_1 L)(t, w(t), \frac{d}{dt}w(t)) = 0,$$

where $\partial_i L$ is the function which is the partial derivative of the function L with respect to the ith argument.[4]

Two different notions of derivative appear in this expression. The functions $\partial_2 L$ and $\partial_1 L$, constructed from the Lagrangian L, have the same arguments as L. The derivative d/dt is an expression derivative. It applies to an expression that involves the variable t and it gives the rate of change of the value of the expression as the value of the variable t is varied.

These are both useful interpretations of the idea of a derivative. But functions give us more power. There are many equivalent ways to write expressions that compute the same value. For example $1/(1/r_1 + 1/r_2) = (r_1 r_2)/(r_1 + r_2)$. These expressions compute the same function of the two variables r_1 and r_2. The first expression fails if $r_1 = 0$ but the second one gives the right value of the function. If we abstract the function, say as $\Pi(r_1, r_2)$, we can ignore the details of how it is computed. The ideas become clearer because they do not depend on the detailed shape of the expressions.

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[3] That the symbols q and $\dot{q}$ can be replaced by other arbitrarily chosen non-conflicting symbols without changing the meaning of the expression tells us that the partial derivative symbol is a logical quantifier, like forall and exists ($\forall$ and $\exists$).

[4] The argument positions of the Lagrangian are indicated by indices starting with zero for the time argument.